TASK – 8

CODE :

```
import math

# 1. Network input values
TOTAL_CHANNELS = 336
AREA = 12
SUBSCRIBERS = [1000, 10000, 20000, 100000]
N_VALUES = [3, 4, 7, 12]
PATH_LOSS = [2.5, 3, 4]

# 2. Calculate Erlang-B blocking probability
def erlang_b(channels, traffic):
    b = 1.0
    for i in range(1, channels + 1):
        b = traffic * b / (i + traffic * b)
    return b

# 3. Find maximum traffic for 2% blocking
def max_traffic(channels):
    low = 0
    high = channels * 3

    for _ in range(100):
        mid = (low + high) / 2

        if erlang_b(channels, mid) > 0.02:
            high = mid
        else:
            low = mid

    return (low + high) / 2

# 4. Calculate hexagonal cell area
def cell_area(radius):
    return (3 * math.sqrt(3) / 2) * radius ** 2

# 5. Calculate Signal-to-Interference Ratio
def calculate_sir(n, cluster):
    sir = (3 * cluster) ** (n / 2) / 6
    return 10 * math.log10(sir)
```

```
for N in N_VALUES:
    channels = TOTAL_CHANNELS // N
    capacity = max_traffic(channels)

    print("\nCluster size:", N)
    print("Channels per cell:", channels)

    for n in PATH_LOSS:
        sir = calculate_sir(n, N)
        print("Path loss:", n,
              "SIR:", round(sir, 2), "dB")

    print("Traffic capacity per cell:",
          round(capacity, 2), "Erlangs")

    for users in SUBSCRIBERS:
        traffic = users * 0.025
        print("Subscribers:", users,
              "Traffic:", round(traffic, 2),
              "Erlangs")

# 7. Calculate initial number of cells
radius = 0.19
cells = math.ceil(AREA / cell_area(radius))

print("\nCELL SPLITTING RESULTS")
print("-------------------------------")
print("Initial cells:", cells)

# 8. Perform three cell-splitting rounds
for k in range(1, 4):
    cells = cells * 4
    radius = radius / 2

    print("Round:", k,
          "Cells:", cells,
          "Radius:", round(radius, 4), "km")

print("\nPROGRAM EXECUTED SUCCESSFULLY")
```

OUTPUT :

```
work.py
CELLULAR NETWORK SIMULATION
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Cluster size: 3
Channels per cell: 112
Path loss: 2.5 SIR: 4.15 dB
Path loss: 3 SIR: 6.53 dB
Path loss: 4 SIR: 11.3 dB
Traffic capacity per cell: 99.62 Erlangs
Subscribers: 1000 Traffic: 25.0 Erlangs
Subscribers: 10000 Traffic: 250.0 Erlangs
Subscribers: 20000 Traffic: 500.0 Erlangs
Subscribers: 100000 Traffic: 2500.0 Erlangs

Cluster size: 4
Channels per cell: 84
Path loss: 2.5 SIR: 5.71 dB
Path loss: 3 SIR: 8.41 dB
Path loss: 4 SIR: 13.8 dB
Traffic capacity per cell: 72.53 Erlangs
Subscribers: 1000 Traffic: 25.0 Erlangs
Subscribers: 10000 Traffic: 250.0 Erlangs
Subscribers: 20000 Traffic: 500.0 Erlangs
Subscribers: 100000 Traffic: 2500.0 Erlangs

Cluster size: 7
Channels per cell: 48
Path loss: 2.5 SIR: 8.75 dB
Path loss: 3 SIR: 12.05 dB
Path loss: 4 SIR: 18.66 dB
Traffic capacity per cell: 38.39 Erlangs
Subscribers: 1000 Traffic: 25.0 Erlangs
Subscribers: 10000 Traffic: 250.0 Erlangs
Subscribers: 20000 Traffic: 500.0 Erlangs
Subscribers: 100000 Traffic: 2500.0 Erlangs
```

```
Cluster size: 7
Channels per cell: 48
Path loss: 2.5 SIR: 8.75 dB
Path loss: 3 SIR: 12.05 dB
Path loss: 4 SIR: 18.66 dB
Traffic capacity per cell: 38.39 Erlangs
Subscribers: 1000 Traffic: 25.0 Erlangs
Subscribers: 10000 Traffic: 250.0 Erlangs
Subscribers: 20000 Traffic: 500.0 Erlangs
Subscribers: 100000 Traffic: 2500.0 Erlangs

Cluster size: 12
Channels per cell: 28
Path loss: 2.5 SIR: 11.67 dB
Path loss: 3 SIR: 15.56 dB
Path loss: 4 SIR: 23.34 dB
Traffic capacity per cell: 20.15 Erlangs
Subscribers: 1000 Traffic: 25.0 Erlangs
Subscribers: 10000 Traffic: 250.0 Erlangs
Subscribers: 20000 Traffic: 500.0 Erlangs
Subscribers: 100000 Traffic: 2500.0 Erlangs

CELL SPLITTING RESULTS
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Initial cells: 128
Round: 1 Cells: 512 Radius: 0.095 km
Round: 2 Cells: 2048 Radius: 0.0475 km
Round: 3 Cells: 8192 Radius: 0.0238 km

PROGRAM EXECUTED SUCCESSFULLY
```